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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

CSC 309 — Past Questions & Materials
by Renance

Course: CSC 309
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 300

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Paper 1 of 1 — CSC 309/414 ARTIFICIAL INTELLIGENCE 2024/2025

Level 300 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
 Department of Computer Science
 1st Semester: BSc (Computer Science Examination) 2022/2023 Session
CSC414 - ARTIFICIAL INTELLIGENCE
 Time Allowed: 2½ Hrs
 Instruction: Attempt Questions 1, 2 AND ANY OTHER 2

Question 1:
 In FUPRE, the probability that a person has HIV is 0.21. However, candidate decides on further blood investigation which shows test was positive with a probability of 0.83. However, when test showed negative - it came with a probability of 0.15. Apply Bayes theorem in evaluating the probability that:

- Find the value Probability of test being positive $P(P)$ and test negative $P(\neg P)$ 5mks
- If the test is positive, show the probability that patient has the virus $\rightarrow P(H|P)$ 5mks
- If the test is positive, the probability that the patient does not have the virus $\rightarrow P(\neg H|P)$ 5mks
- If the test is negative, the probability that the patient has the virus $\rightarrow P(H|\neg P)$ 5mks
- If the test is negative, the probability that the patient does not have the virus $\rightarrow P(\neg H|\neg P)$ 5mks

Question 2:

- Describe the Concept of Genetic Algorithm 2mks
- In a problem space of chromosomes that are properly initialized, the fitness function ranges from [0.65, 0.9]. We discovered 2-candidates with decimal values [107 and 181] which are selected for reproduction. Work out the following:
 - Perform the Binary encoding of the data therein 2mks
 - Show the working using a 1-point, 2-point and multi-point crossover 6mks
- Is it possible for the solutions to alter? If Yes, Why and state the process responsible. 4mks
- How does GA deal with the process mentioned above in (c) if it is possible and when it occurs 3mks
- Give any 8-configurations of the 8-Queen problem with an appropriate diagram 8mks

Question 3:

- Describe in your words: (1) Mixed Strategy, (2) Minimax, (c) Maximin, (d) Pure strategy 4mks
- Describe the typical prisoner's dilemma 2mks
- State the Expected Payoff theorem 2mks
- Given the table below, describe the saddle-point in the game 2mks

		J			
		8	10	17	9
		9	2	1	10
		6	15	8	7
		10	11	15	13
I	P	8	10	17	9
	Q	9	2	1	10
	R	6	15	8	7
	$\neg R \wedge \neg Q$	10	11	15	13
	$R \vee \neg Q$				

Question 4:

- Given P, Q and R, evaluate the implication for the falsity or truth of the argument 10mks

P	Q	R	$\neg R \wedge \neg Q$	$R \vee \neg Q$	$\neg P \vee R$	$\neg P \wedge R$	$R \wedge P$	$\neg Q \rightarrow R$	$Q \rightarrow P$	$R \rightarrow Q$	$P \leftrightarrow R$	$\neg Q \leftrightarrow R$
T	F	T										
F	T	T										
F	T	F										
T	T	T										
F	F	F										
F	F	T										
T	T	F										
T	F	F										

Question 5: Given the roll of two dice, evaluate the following

- Draw the Probability Matrix table for the roll of two dice 4mks
- $P(\text{total of an 11 AND roll of just first die, it yields a 3})$ 2mks
- $P(\text{total of 8 OR total of 6})$ 2mks
- Describe the term: (1) Modus Ponens, (b) Modus Tollens 2mks

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